LEC 2: DT SYSTEMS OVERVIEW

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Courtesy: Dr. Khurram Ali

Chapter 2: Overview (SNS Revision)

- Discrete Time Signals: Review
- Discrete Time systems: Review
- Analysis of Discrete Time LTI systems: Review
- Description of Discrete time systems: concept of difference equation.

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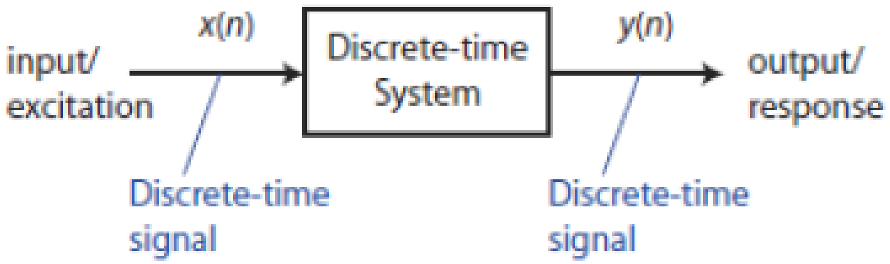
Discrete Time Systems: Overview

A discrete time system is a device that operates on one discrete time signal called input and produces another discrete time signal called output



- Input-output description (exact structure of system is unknown or ignored):

$$y(n) = \mathcal{T}[x(n)]$$

- "black box" representation:

$$x(n) \xrightarrow{\mathcal{T}} y(n)$$

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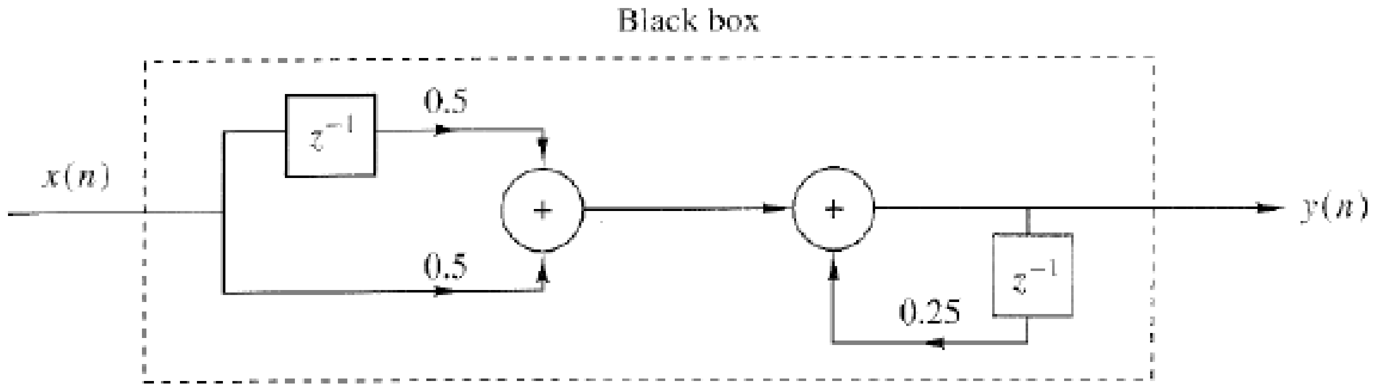
Block Diagram Representation

We can express any relation between input and output of a discrete time system by an equivalent block diagram

$$y(n) = \frac{1}{4}y(n-1) + \frac{1}{2}x(n) + \frac{1}{2}x(n-1)$$



We will come to it in detail later in this lecture !

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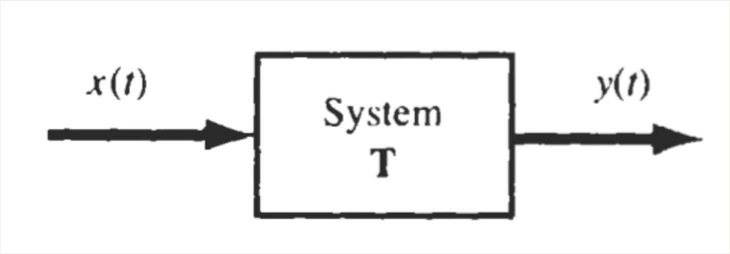
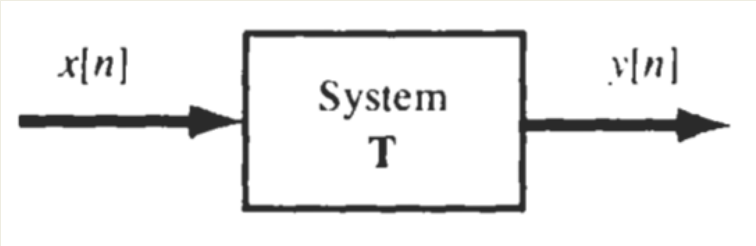
Types of Systems

- Continuous Time System
  - Systems in which input and output, both signals are continuous in nature.
  - $x(t) \rightarrow y(t)$
  - e.g. Audio and video amplifier, power supplies etc.

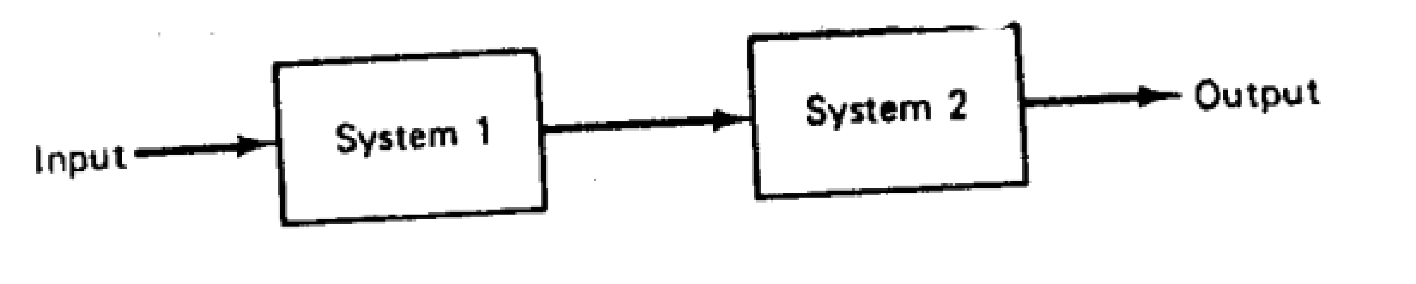


- Discrete Time System
  - Systems in which input and output, both signals are discrete in nature.
  - $x[n] \rightarrow y[n]$
  - e.g. microprocessors, shift registers

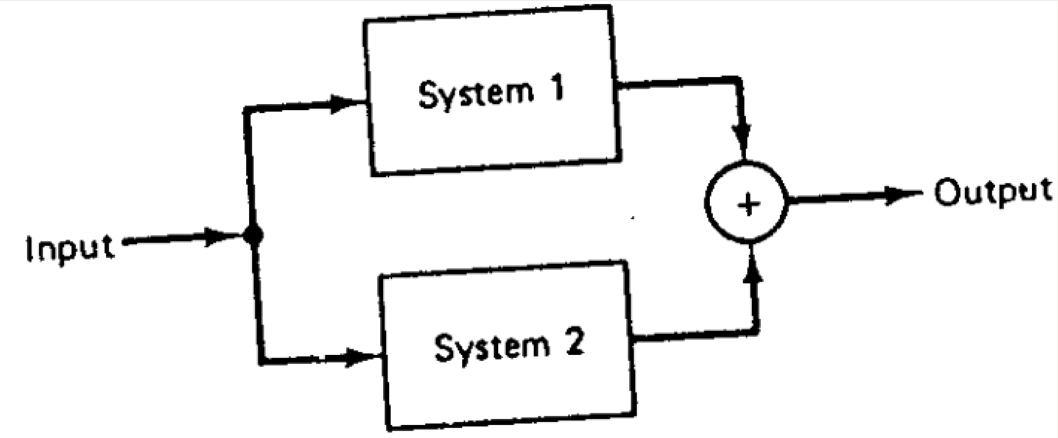


Interconnection of Systems

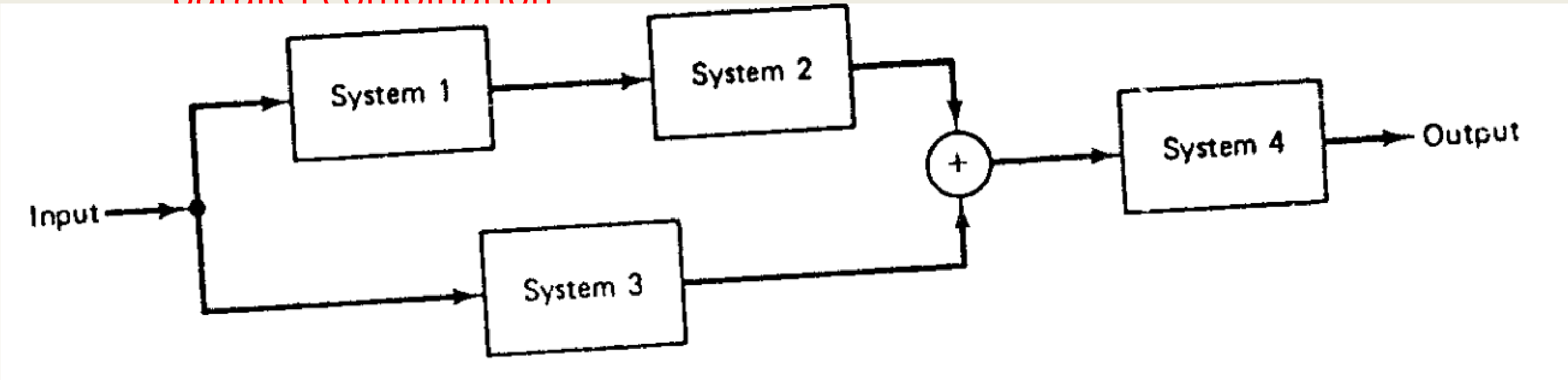
- Systems can be interconnected in multiple ways
    - 1) Series(Cascaded) Interconnection
        - Output of system 1 is input of other system
        - Overall impulse response is equal to convolution of individual impulse responses.



- 2) Parallel Interconnection
  - Same input is applied to all systems.
  - Overall impulse response is equal to addition of individual impulse responses

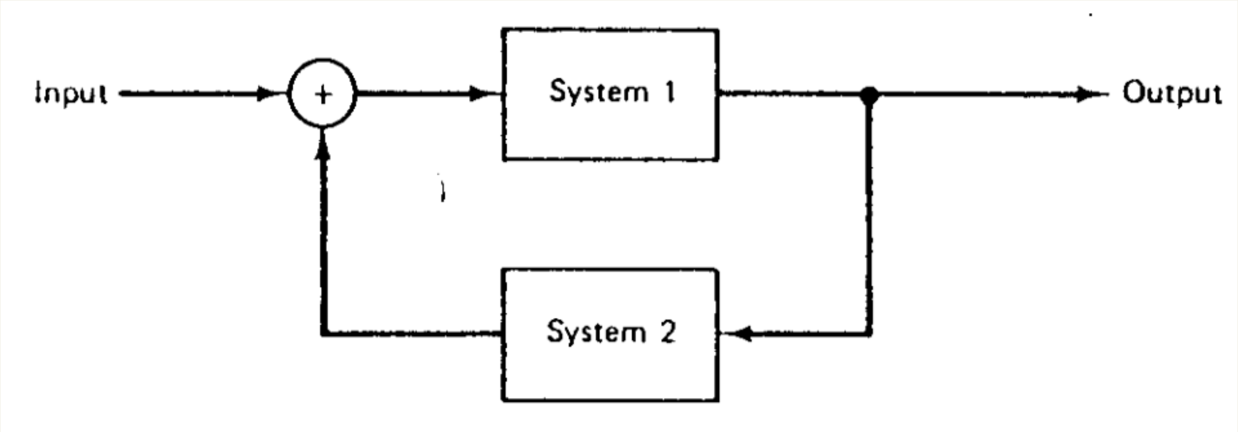


- 3) Series/Parallel Interconnection
  - Combination of series and parallel combination

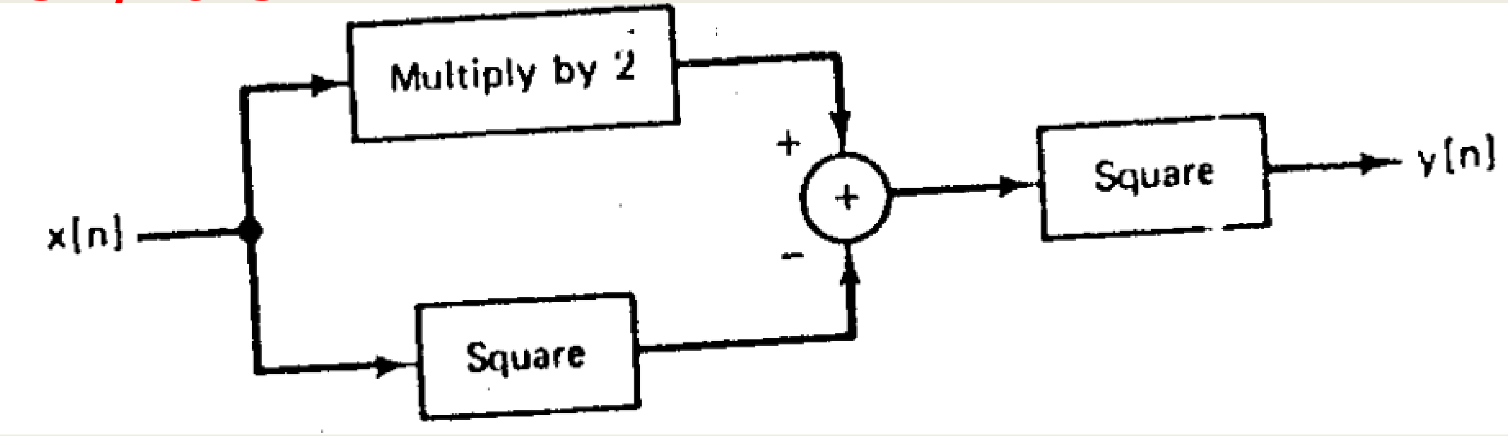


4) Feedback Interconnection

- Output of system 1 is the input to system 2, while the output of system 2 is fed back and added to the external input to produce the actual input to system 1.



Exp 7.1: Write relationship between input and output signals



- Multiple input $x[n]$ by 2 → $2x\,[n]$
- Square of input $x[n]$ → $x[n]^2$
- Relationship is: $y\,[n] = \left(2x\,[n] - x[n]^2\right)^2$

Discrete Time Systems: Classifications

Common System Properties:

static	vs.	dynamic
time-invariant	vs.	time-variant
linear	vs.	nonlinear
causal	vs.	non-causal
stable	vs.	unstable systems
⋮		⋮

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1) Causal and Non-Causal Systems

- Response of the causal system to an input does not depend on future values of that input but depends only on the present and past values.

- **Exp1:** $\boldsymbol{y(n) = x(n^2)}$
- **Solution:**
- Put different values of n in above equation to check whether it is causal or not
    - Put $n = 0$
        - $y(0) = x(0^2) \Rrightarrow y(0) = x(0)$
    - Put $n = 1$
        - $y(1) = x(1^2) \Rrightarrow y(1) = x(1)$
    - Put n$= -1$
        - $y(-1) = x(-1^2) \Rrightarrow y(-1) = x(1)$

- $y(n)$ is depending on previous and present values of $x(n)$ only so it is a non-causal system

- **Exp2:** $\boldsymbol{y(n) = 2x(n-1)}$
- **Solution:**
- Put different values of n in above equation to check whether it is causal or not
  - Put $n = 0$
    - $y(0) = 2x(0-1) \Rrightarrow y(0) = 2x(-1)$
  - Put $n = 1$
    - $y(1) = 2x(1-1) \Rrightarrow y(1) = 2x(0)$
  - Put $\mathrm{n} = -1$
    - $y(-1) = 2x(-1-1) \Rrightarrow y(-1) = 2x(-2)$
- $y(n)$ is depending on previous and present values of $x(n)$ only so it is a causal system

2) Static (memoryless) and Dynamic (with memory) Systems

- Static systems are also known as memoryless systems
- Static systems contain no storage elements (thus, no integrals, derivatives or signal delays)
- A static or memoryless system is a system with an output signal whose values depends upon the present value of the input signal *only*. Otherwise the system is dynamic or with memory.

- **(ii)** $\boldsymbol{y(n) = x(n) - x(n-1)}$
- **Solution:**
- Put different values of n in above equation to check whether it is static or not
  - Put n$= 0$
    - $y(0) = x(0) - x(0-1) \Rrightarrow y(0) = x(0) - x(-1)$
  - Put $\mathrm{n} = 1$
    - $y(1) = x(1) - x(1-1) \Rrightarrow y(1) = x(1) - x(0)$
  - Put $n = -1$
    - $y(-1) = x(-1) - x(-1-1) \Rrightarrow y(-1) = x(-1) - x(-2)$
- $y(n)$ is depending on previous and present of $x(n)$ values only so it is a dynamic system

3) Time Invariant and Time Variant Systems

- A system is **time invariant** if the time shift in the input signal results in corresponding time shift in the output.
- Let $y(n) = f[x(n)]$ i.e. $y(n)$ is response of $x(n)$. Then if $x(n)$ is delayed by time n_1 then output $y(n)$ will also be delayed by the same time. i.e.
- $f\ [x(n-n_1)] = y(n-n_1)$

Steps to test for time invariance property

- Step 1: Determine the output of system for delayed input i.e. $x(n-n_1)$
  - $y(n, n_1) = f[x(n-n_1)]$

- Step 2: Then delay the output itself by n_1 i.e. $y(n-n_1)$

- Step 3: If
  - $y(n, n_1) \neq y(n-n_1)$ → Time variant
  - $y(n, n_1) = y(n-n_1)$ → Time invariant

- $(i)\ y(n) = nx(n)$
- Step 1: Determine the output for delayed input i.e. . $x(n - n_1)$
  - $y(n, n_1) = f[x(n - n_1)]$
  - $y(n, n_1) = nx(n - n_1)$
- Step 2: Now delay the output by n_1. Hence we have to replace n by $n - n_1$
  - $y(n - n_1) = (n - n_1)x(n - n_1)$
- Step 3: Compare $y(n, n_1)$ with $y(n - n_1)$
  - $y(n, n_1) \neq y(n - n_1)$
- Both are not same so system is *time variant*

4) Linear and Non-Linear Systems

- A system is said to be linear if it follows the superposition principle.
- Consider two systems defined as
  - $y_1(n) = f\,[x_1(n)]$ i.e. $x_1(n)$ is input and $y_1(n)$ is output
  - $y_2(n) = f\,[x_2(n)]$ i.e. $x_2(n)$ is input and $y_2(n)$ is output
- Discrete time system is said to be linear if
  - $f[a_1\, x_1(n) + a_2\, x_2(n)] = a_1\, y_1(n) + a_2\, y_2(n)$

If $y_3(t) = y_3^{'}(t)$ → Linear

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Since $y_3(t) = y_3'(t)$ → Linear

5) Stable and Unstable Systems

Bounded Input

Bounded Output

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Thank You !!!

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